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Factors driving young users' engagement with Facebook: Evidence from Brazil

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ABSTRACT

User engagement has recently been the focus of attention for marketing planners who want to capture the enormous opportunities provided by social media. In this study, we investigate the drivers of social media user engagement by extending an existing model of social media participation by Cheung, Chiu and Lee (2011). We included in the model three social influence factors, five uses and gratifications factors, and one social presence factor. Results from an online survey with a sample of young Facebook users in Brazil ($n = 1126$) reveal four significant factors, i.e. a) subjective norm b) social identity, c) entertainment value, and d) maintaining interpersonal interconnectivity. Our empirical results also show an improved model fit over the existing model of user participation used by Cheung et al. (2011). This research adds significance to the literature by extending an existing theoretical model to study a new phenomenon in a different cultural context. We also provide practical recommendations for managing brand fan pages on how to foster user engagement with social media.

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1. Introduction

User engagement has been recognised as a prerequisite for the success of virtual environments including social media (Verhagen, Swen, Feldberg, & Merikivi, 2015), and recently there is growing body of research examining user engagement particularly among students and young people (e.g. Byun & Loh, 2015; Cruz-Benito, Therón, García-Peñalvo, & Pizarro Lucas, 2015; Habibi, Laroche, & Richard, 2014; Lim, Hwang, Kim, & Biocca, 2015; Pellas, 2014; Wiebe, Lamb, Hardy, & Sharek, 2014). However, studies examining the drivers of user engagement with social media are still limited. One exception is Verhagen et al. (2015), who use the theory of uses and gratification to link the characteristics of virtual environments, and perceived benefits of using these environments with user engagement intentions. In contrast, prior studies on motivation of social media participation have employed a combination of various theories such as 'social influence', 'uses and gratification', and 'social presence' (e.g. Cheung, Chiu & Lee, 2011; Dholakia, Bagozzi, & Pearo, 2004).

There is evidence about the influence of sociocultural factors on the design of social networking site (SNS, K.-S. Shen, 2013). However, there is also a lack of studies about the use of SNSs in cultures other than Asia and North America (Cho & Park, 2013). Yet, cultural differences could have considerable influence on social media users' attitudes and communication style (Cho & Park, 2013), and possibly on motivations to engage with social media. For this reason, knowledge on the specificities regarding other cultures would be critical in explaining the participant's engagement in SNSs. In the scope of this research, focussing on users in Brazil would bring interesting reflections for theoretical development about factors affecting use of Facebook in different cultural contexts. The results reported by two separate studies we previously cited show contrasting results: Dholakia et al. (2004) report that their model explained 54% of the variance of its dependent variable using the data of user participation in virtual communities in the US. Cheung et al. (2011) included an additional construct (social presence) to the model using the data of a sample of Facebook users in Hong Kong, yet the explanatory power of their model fell considerably, explaining only 28% of the variance of the dependent variable. The difference in the models' explanatory powers may be attributed to two possible reasons: a) different characteristics related social media investigated in the two studies, i.e. between general virtual communities and Facebook; b) different cultural

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contexts of the two studies, one in the US and the other Hong Kong. According to [The Hofstede Centre \(2015\)](#), Brazil ranks higher in the scores of Uncertainty Avoidance (76), Indulgence (59) and Individualism (38) than Hong Kong (29, 17 and 25, respectively), but lower in the term of Long Term Orientation (44 versus 61). Brazil also ranks very differently from the US on all the dimensions of Hofstede's culture framework. Thus further research of the conceptual model in a new context such as Brazil is required to uncover interesting reflections for theoretical development about factors affecting user engagement with social media.

This research addresses the research gap by extending an existing conceptual model of social media participation ([Cheung et al., 2011](#); [Dholakia et al., 2004](#)) to study social media engagement in a new and important cultural context. Our study makes three important contributions to the literature. First, our investigation provides empirical evidence of extending the research model first developed by [Dholakia et al. \(2004\)](#) and later modified by [Cheung et al. \(2011\)](#) to study a new construct (user engagement) in a new and important cultural context (Brazil). Our results show an improved model fit than that reported in the previous study ([Cheung et al., 2011](#)). Second, by integrating 'social influence', 'uses and gratifications', and 'social presence', we reveal the unique driving factors of social media engagement, thus improving the current understanding of the key drivers of social media user engagement, extending the study by [Verhagen et al. \(2015\)](#). Third, our research has practical implications for companies wishing to foster user engagement with their brands through social network sites.

We test our hypotheses using survey data collected from a sample of young Facebook users in Brazil. In addition to the cultural differences from the commonly researched contexts such as North America and Asia, Brazil is an especially appropriate and important field context for testing the conceptual model, because Brazil is a country strongly influenced by social networking ([Mintel, 2014](#)). Statistics in April 2014 show that the country has the third largest number of Facebook users in the world, just behind the US and India, and there are more than 86 million Facebook users in Brazil, a penetration of 84% amongst the country's Internet users ([Mintel, 2014](#); [Socialbakers, 2014](#)). Facebook in Brazil is not just a social networking site, it is an instrumental tool for nearly every aspect of life in Brazil, including e-commerce transactions which are increasing rapidly ([Mintel, 2014](#)). Mintel's research suggests that about 15% of users who have clicked on Facebook advertisements have purchased at least once online in the past 12 months, compared to just 8% for the total population. For the consumers aged 16–24 years – the largest age group of Facebook users in Brazil ([Socialbakers, 2014](#)), the percentage reached a staggering 28% ([Mintel, 2014](#)). As argued in Mintel's report, Brazil is leading the way into the future of e-commerce on Facebook, and other countries could learn a lot from the Brazilian experience. Thus, understanding what drives young Brazilian users' engagement with Facebook is important not only for scholarship but also for social media marketing practice in Brazil and beyond.

2. Theory and hypotheses

Research into the use of social media including social networking sites (SNSs) such as Facebook and Twitter has drawn upon a wide variety of sources ranging from personal behaviour theories, social behaviour theories, to mass communication theories. A recent literature review by [Ngai, Tao, and Moon \(2015\)](#) show that these include: 15 theories related to personal behaviour, such as personality traits theory, technology acceptance model, and the theory of planned behaviour; 13 theories related to social behaviour, such as social capital theory, social exchange

theory, and social influence theory; and 3 theories of mass communication such as media richness theory, and uses and gratifications theory (U&G). Social media usage intention and behaviour are among the major outcome variables investigated in the literature (e.g. [Fischer & Reuber, 2011](#); [Parra-López, Bulchand-Gidumal, Gutiérrez-Taño, & Díaz-Armas, 2011](#); [Zhong, Hardin, & Sun, 2011](#)). User engagement is one of the key factors for the success of the information system ([Hwang & Thorn, 1999](#)) and potentially is also the case for social media. Recent research in marketing shows that engaged consumers exhibit high levels of satisfaction, loyalty, connection and emotional bonding with a specific object, such as a brand or a social networking site ([Brodie, Ilic, Juric, & Hollebeek, 2013](#)). In our attempt to identify the key factors that influence user engagement, we first review the conceptualisation of the user engagement construct, followed by a review of social influence theory, U&G, social presence theory and relevant research to develop our hypotheses.

2.1. Conceptualisation of user engagement

Being engaged 'is to be involved, occupied, and interested in something' ([Higgins, 2006](#), p. 442). [Hwang and Thorn \(1999\)](#) use 'engagement' as a general term that refers to both user involvement and participation in an information system, whereas there has been a proliferation of definitions of engagement proposed in the literature in recent years. For example, [Porter, Donthu, MacElroy, and Wydra \(2011\)](#) focus on behaviour that reflects online community members' willingness to participate and cooperate with other members, while [Van Doorn et al. \(2010\)](#) propose five dimensions of customer engagement, i.e. valence, form or modality, scope, nature of engagement's impact, and customer goals. [Mollen and Wilson \(2010\)](#) define online engagement as 'a cognitive and affective commitment to an active relationship with the brand as personified by the website' (p.923). [Brodie et al. \(2013\)](#) suggest that engagement is a context-dependent, psychological state characterized by fluctuating intensity levels, but they consider it as a multidimensional concept comprising cognitive, emotional, and/or behavioural dimensions. In contrast, [Calder and Malthouse \(2008\)](#) build on [Higgins' \(2006\)](#) conceptualisation of the term and emphasise that engagement is a state of involvement and connectedness between the user and the engagement object (for example, the media), and they believe that engagement is a motivational force to make something happen, while the actual behaviour is the consequence of engagement, not engagement itself. Thus in this study, we focus on the behavioural intention aspect of user engagement to examine young Brazilian users' intentions to be involved with the participation and socialization experience in Facebook ([Calder & Malthouse, 2008](#)).

2.2. Theory of social influence

The theory of social influence explains the effect of others on the individual's behaviour ([Kelman, 1958](#)). Compliance happens when an individual accepts the social influence to obtain a reward, support or approval from significant others, avoiding punishment. Internalisation occurs when one finds that he or she share the values of other group members ([Bagozzi & Dholakia, 2002](#)). Identification happens when an individual under social influence attempts to establish and maintain a satisfying self-defining relationship to another person or group. Prior studies often use subjective norm, group norm and social identity to reflect these social influence processes (e.g. [Bagozzi & Lee, 2002](#); [Cheung et al., 2011](#); [Dholakia et al., 2004](#); X.-L. Shen, Cheung, & Lee, 2013; [Venkatesh & Davis, 2000](#)).

Subjective norm refers to an individual belief that people who are important to her/him thinks she/he should perform the behaviour in question (Fishbein & Ajzen, 1975; Venkatesh & Davis, 2000). It shows the influence of expectations of particular others, indicating social normative compliance (Fishbein & Ajzen, 1975). Several studies have used the concept of subjective norm to evaluate the intention to adopt social networks (e.g. S.-C. Chen, Yen, & Hwang, 2012), whereas Cheung et al. (2011) report that subjective norm was not a significant factor in driving Facebook use intention. We postulate the following hypothesis for further testing:

H1. Subjective norm positively influences Facebook engagement among young Brazilian users.

Group norm is based on the processes of internalization and translated into shared values or goals (Bagozzi & Lee, 2002; Eagly & Chaiken, 1993). In other words, group norm is related to the use of common self-guides to achieve one's own goals that are shared with peers (Dholakia et al., 2004). Internalization processes also lead to feelings of moral obligation in relation to the welfare of other members of the group and it goes beyond offering favours as beneficial actions on behalf of the group and its members (Tyler, 1999). In a social network, a participant assembles a group of friends and shares with them important things in their life, especially when there is the perception that this group shares similar goals and objectives (Christian, Bagozzi, Abrams, & Rosenthal, 2012). Thus:

H2. Group norm positively influences Facebook engagement among young Brazilian users.

Social identity refers to the way a person thinks about herself/himself based on their social groupings (Hannum, 2007). An individual normally develops a social identity based on her/his needs of belonging to a social group, which brings to her/him emotional and self-enhancement value significance (Bagozzi & Lee, 2002; Tajfel, 1982). The sense of belonging is one of the major benefits obtained through participation in group discussion via social media. Examining Facebook use among a sample of students in United States, Lin, Fan, and Chau (2014) have recently reported that sense of belonging positively influenced intention to continue using Facebook. Although Cheung et al. (2011) found that social identity did not have a significant impact on Facebook use, several other studies found that social identity has a significant impact on online community or SNS participation (e.g. Hsu & Lin, 2008; Li, 2011; X.-L. Shen et al., 2013). Thus:

H3. Social identity positively influences Facebook engagement among young Brazilian users.

2.3. Theory of uses and gratifications

The theory of uses and gratifications (U&G) originated in the 1940s, and is frequently used today to understand consumer motivations to use online SNS (e.g. G. M. Chen, 2011; Dholakia et al., 2004). The U&G assumes that individuals actively use media to fulfil their particular needs. Park, Kee, and Valenzuela (2009) reveal four primary needs for participating in groups within Facebook: socializing, entertainment, self-status seeking, and information seeking. Pai and Arnott (2013) identify four main values derived from SNS use: belonging, hedonism, self-esteem, and reciprocity. Dholakia et al. (2004) categorise the fulfilment of different needs into five values: purposive value, self-discovery, social enhancement value, maintaining interpersonal connectivity, and entertainment value.

Purposive value refers to informational and instrumental values derived from performing some predetermined instrumental purpose, including giving or receiving information through

participation in virtual communities (Dholakia et al., 2004). Park et al. (2009) indicate that one of the reasons for participating in Facebook Groups among college students is to obtain information about events, products and services. In Cheung et al.'s (2011) study, the impact of purposive value on Facebook use intention was not significant, while Lampe, Wash, Velasquez, and Ozkaya (2010) found that although seeking information was not significantly associated with social media use frequency, it was significantly associated with future use intention. Thus, we put the following hypothesis to further test:

H4. Purposive value positively influences Facebook engagement among young Brazilian users.

Self-discovery refers to the understanding and deepening salient aspects of the self through social interactions, the ability to help the person form, clearly define and develop their own preferences, tastes and values (Dholakia et al., 2004). In a study of intention to continue using a virtual community in China (Jin, Cheung, Lee, & Chen, 2007), self-discovery value was found to significantly affect a user's continuance participation in the virtual community mediated by sense of belonging. However the findings reported in the literature are not consistent, for example, Cheung et al. (2011) and Lampe et al. (2010) report that self-discovery was not a significant factor for social media use. Thus we put the hypothesis for a further test:

H5. Self-discovery positively influences Facebook engagement among young Brazilian users.

Maintaining interpersonal connectivity refers to the social benefits derived from establishing and maintaining contact with other people, such as social support, friendship and intimacy. Facebook and social media in general enable users to share content, communicate with each other, and stay connected with friends. Cheung et al. (2011) show a positive, although somewhat weak, link between maintaining interpersonal connectivity and Facebook use intention among Hong Kong Facebook users. While Lin et al. (2014) show a positive influence of Facebook connectedness on user satisfaction and sense of belonging and G. M. Chen (2011) found that gratification from connecting with others is positively associated with Twitter use frequency. We therefore postulate that:

H6. Maintaining interpersonal interconnectivity positively influences Facebook engagement among young Brazilian users.

Social enhancement value is derived from gaining acceptance and approval of the other participants, improving her/his social status within the online community because of his/her contribution (Dholakia et al., 2004). Social enhancement value is attained from fulfilling user self-status seeking need when using social media (Park et al., 2009). Earlier research by Flanagin and Metzger (2001) show that social enhancement is one of the motivations for Internet use. Cheung et al. (2011) found that social enhancement was a significant factor driving Facebook use. More recent research by Pai and Arnott (2013) reveal that SNS users gain self-publicity, peer respect from having a unique user profiles and popularity appreciated by friends on social media. Given the empirical evidence of the social enhancement benefits for SNS users, we hypothesize that:

H7. Social enhancement positively influences Facebook engagement among young Brazilian users.

Entertainment value is the perceived degree of enjoyment, fun or feelings of relaxation derived from using an information system or SNS (Dholakia et al., 2004; Verhagen, Feldberg, van den Hooff, Meents, & Merikivi, 2011). Similar terms used in the information management literature include 'hedonic gratification' (Xu, Ryan, Prybutok, & Wen, 2012) and 'perceived playfulness' (Ahn, Ryu, & Han, 2007; Chung & Tan, 2004). Verhagen et al. (2011) argue that

entertainment value is an intrinsic value as it offers instant pleasure, irrespective of the rational and goal-oriented tasks. Prior research has shown pleasure has a significant impact on user satisfaction with and commitment to SNS or the 'virtual world' (e.g. Cheung et al., 2011; Lin et al., 2014; Verhagen et al., 2011; Xu et al., 2012). Thus:

H8. Entertainment value positively influences Facebook engagement among young Brazilian users.

2.4. Theory of social presence

Social presence is a central element of the social context (Short, Williams, & Christie, 1976) and has been used extensively in computer-mediated communication research (Weisberg, Te'eni, & Arman, 2011). Short et al. (1976, p. 65) define social presence of a particular communication medium as "the degree of salience of the other person in the interaction and the consequent salience of the interpersonal relationships," and recent definitions of social presences have been extended to the degree of positive interpersonal and emotional connection between users of online media (Cui, Lockee, & Meng, 2013). The theory argues that social presence affects medium use (Weisberg et al., 2011). Studying user perception of the social presence of an ecommerce website, Gefen and Straub (2004) found that social presence influences trust, especially benevolence, and contributes to online purchase intentions. The study by Cheung et al. (2011) shows that social presence has the strongest impact on Facebook use among students in Hong Kong. Thus:

H9. Social presence positively influences Facebook engagement among young Brazilian users.

3. Methods

To collect the empirical data of this study, we use an online survey. We used two items to measure Facebook user engagement intention adapted from Calder and Malthouse (2008), and the measures of remaining constructs were adopted from Cheung et al. (2011) and Dholakia et al. (2004) in developing our questionnaire. To reach young participants across the country, we chose the largest company that offers job openings for students and trainees in the Brazilian market. There are approximately 50 thousand unique users accessing this site daily, and there are about 3.8 million young people registered. Soon after the users login on the site, we invited the users to answer our questionnaire voluntarily. We received 5655 responses and randomly extracted 20% for the sample for data analysis (i.e. $n = 1126$). The average age of the sample is 21 ($SD = 6.37$) and most respondents are aged 16–24 (74% of the sample). This age range is comparable to the research by Lin et al. (2014) who had 82% of respondents in their sample between 20 and 23 years old. There are slightly more females (57.6%) than males. Respondents are mostly students from two levels of education: 34.1% of students from high school and 58.1% are students of higher education. The profile closely matches the official Brazilian government data (IBGE., 2013), which shows that there are 34.1 million young people between 15 and 24 years old, which is 20.1% of the total population, roughly equally split in terms of gender composition (51.3% females). Additionally, among young people between 16 and 24 years, 46.9% have some sort of employment.

4. Results

We used partial least square structural equation modelling (PLS-SEM) to perform our data analysis. PLS-SEM has several advantages

over covariance-based SEM. For example, it works well when the research model has many indicators and relationships are estimated. PLS-SEM is also particularly well suited to settings where the emphasis is on predictive modelling (Hair, Ringle, & Sarstedt, 2011). The software we used is SmartPLS 2.0 M3 (Ringle, Wende, & Will, 2005). Following Hair et al. (2011), we computed the t -values using 5000 bootstrap samples to establish the significance value for each path coefficient.

4.1. Measurement model assessment

First, we assessed the measurement model by examining the item loading, composite reliability, convergent validity, and discriminant validity. As shown in the Table 1, most of the item loadings are above the recommended 0.7 (Hair et al., 2011) with a few exceptions lower than 0.7 which were removed from further analysis (EV2, PV2, PV7, PV8, PV9, and SP2, shown in italic in Table 1). All factor loadings are significant, and the composite reliability (CR) exceeds the recommended level of 0.7, and the average variance extracted (AVE) values are above the recommended level of 0.5 (Hair et al., 2011).

The factor loadings (and cross loadings) of all indicators to their respective construct are greater than their loadings on other constructs (Appendix A). We then compared the square roots of the AVE with the latent variable correlations and the results show good discriminant validity (Table 2). All square roots of each latent variable's AVE are greater than the latent variable's correlations with any other construct in the model, each pair of latent variables meet the criterion of Fornell and Lacker (1981) in support of discriminant validity.

4.2. Hypothesis test

Table 3 shows the results of the PLS-SEM estimation with the path coefficient values and their associated t -values of the paths. The model accounts for 42 per cent of the variation in user engagement. The significant and positive paths are from social norm (H1), social identity (H3), maintaining interpersonal interconnectivity (H6) and entertainment value (H8) to user engagement. The remaining paths are either insignificant (group norm, purposive value, self-discovery and social presence) or not positive (social enhancement). Subjective norm is the most significant predictor of user engagement, followed by social identity.

5. Discussion and conclusions

The main objective of this study was to identify the factors influencing young Brazilians' engagement with Facebook, by extending an existing model of social media participation integrating three well-established theories, i.e. social influence, uses & gratifications, and social presence. Our empirical study shows: a) the unique features of Facebook user engagement among young Brazilians, and b) an improved model fit over the existing model of user participation (Cheung et al., 2011). This research has implications for both theories and practice.

5.1. Theoretical implications

Among the three factors of social influence evaluated in this study, subjective norm and social identity were the significant factors explaining young Brazilians' engagement with Facebook. This indicates that the processes of influence occurred in two forms of social influence, i.e. compliance and identification (Kelman, 1958). The significant impact of subjective norm suggests that individuals perceive that other Facebook users want him/her to

Table 1
Scale items and convergent validity.

Constructs/items	AVE	CR	Loading
Entertainment value (EV)	0.72	0.88	
EV1 To have fun			0.87
EV2 To play			0.45
EV3 To relax			0.84
EV4 To pass the time when I'm bored			0.83
Group norm (GN)	0.88	0.94	
GN1 What is your effort to meet this goal?			0.93
GN2 What is the average effort of the "Group of Friends" to meet this goal?			0.95
User Engagement Intention (INT)	0.88	0.94	
INT1 I would actively contribute to the conversation with my friends on Facebook in the next two weeks.			0.94
INT2 I would actively interact with my friends on Facebook in the next two weeks.			0.94
Maintaining interpersonal interconnectivity (MII)	0.76	0.86	
MI1 To have issues in common with others			0.83
MI2 To keep in touch			0.91
Purposive value (PV)	0.64	0.90	
PV1 For information			0.76
PV2 To learn how to do things			0.68
PV3 To provide information to other			0.80
PV4 To contribute to a set of information			0.88
PV5 To generate ideas			0.82
PV6 To make changes			0.73
PV7 To get someone to do something for me			0.62
PV8 To solve problems			0.66
PV9 To make decisions			0.66
Self-discovery (SD)	0.77	0.91	
SD1a To learn more about me			0.87
SD1b To learn more about each other			0.89
SD2 For more insights about me			0.88
Social enhancement (SE)	0.90	0.95	
SE1 To impress			0.96
SE2 To feel important			0.94
Social identity (SI)	0.69	0.93	
SI1 Please indicate to what degree your image coincides with this "Group of Friends"?			0.73
SI2 As you consider the degree of similarity between you and a group of friends are listed when you interact?			0.78
SI3 What is your connection with the "Group of Friends" listed?			0.86
SI4 With what intensity you feel part of their "Group of Friends"?			0.88
SI5 I consider myself a valuable member to this my "Group of Friends".			0.87
SI6 I am an important member for this my "Group of Friends"			0.86
Subjective norm (SN)	0.83	0.91	
SN1 People who influence my behaviour I approve join Facebook.			0.91
SN2 People important to me approve I join Facebook.			0.92
Social presence (SP)	0.67	0.89	
SP1 There is a sense of human contact in Facebook			0.84
SP2 There is a sense of impersonality on Facebook			0.53
SP3 There is a sense of sociability on Facebook			0.85
SP4 There is a feeling of human warmth on Facebook			0.77
SP5 There is a sense of human sensitivity on Facebook			0.81

Notes: AVE = Average variance extracted, CR = composite reliability. Items in italic were excluded from further analysis.

perform a specific behaviour and that they have the ability to reward or to punish his/her behaviour. In other words, user engagement with Facebook in Brazil is influenced by people who are important to them. As suggested by Cheung et al. (2011),

Table 2
Latent variable correlations & square roots of AVE.

	EV	GN	INT	MII	PV	SD	SE	SI	SN	SP
EV	0.85									
GN	0.33	0.94								
INT	0.37	0.37	0.94							
MII	0.58	0.41	0.44	0.87						
PV	0.41	0.4	0.38	0.57	0.80					
SD	0.41	0.29	0.24	0.49	0.5	0.88				
SE	0.43	0.22	0.13	0.39	0.36	0.59	0.95			
SI	0.41	0.56	0.51	0.51	0.47	0.30	0.17	0.83		
SN	0.32	0.31	0.54	0.40	0.37	0.29	0.17	0.45	0.91	
SP	0.48	0.34	0.32	0.51	0.47	0.48	0.43	0.40	0.35	0.82

Notes: Boldface numbers on the diagonal are the square root of the AVE for each construct.

compliance occurs when the user has no experience with a new system and second-hand information is important for his/her usage decisions. Our results corroborate their findings.

The significant impact of social identity denotes that Facebook users accept social influence because they want to establish or maintain a satisfactory relationship with other social network members. As suggested by Cheung et al. (2011), identification occurs when there is self-awareness of one's membership of a group, as well as the emotional and evaluative significance of this membership. To Zaglia (2013), the transference of information made by peers and influencers tend to be more valuable and accepted by members of the same network because they share the same object of interest and affection. The results of this research show that young Brazilian Facebook users seem to have a special preoccupation about their feeling of belonging and this feeling motivates their intention to engage with Facebook. Social needs are, after the physiological needs, the second category of needs that motivates human behaviour (Foxall, Goldsmith, & Brown, 1998). Our findings corroborate the phenomenon reported in Brazil that the social identity needs of young people have not been satisfied within the

Table 3Results of factors explaining Facebook user engagement ($R^2 = 0.42$).

Hypothesis	Predictors	Coefficient	t-value	Supported?
H1	Subjective norm	0.35	9.33**	Yes
H2	Group norm	0.06	1.48	No
H3	Social identity	0.22	5.36**	Yes
H4	Purposive value	0.06	1.64	NS
H5	Self-discovery	−0.02	0.53	NS
H6	Maintaining interpersonal interconnectivity	0.11	2.92**	Yes
H7	Social enhancement	−0.08	2.29*	No
H8	Entertainment value	0.11	3.31**	Yes
H9	Social presence	−0.01	0.04	NS

Note: * $p < 0.05$, ** $p < 0.01$.

primary social system. The Program for International Student Assessment (PISA) reported that in Brazil students' sense of belonging at school deteriorated between 2003 and 2012. In 2003 only 8% of students reported that they felt lonely, this figure more than doubling to 19% in 2012 (PISA, 2012). Given that the motivational process is triggered by unmet needs that arise from the difference between the individuals' actual and aspired states (Urdan & Urdan, 2010), the results of our research suggest that Facebook is probably used by young people in Brazil to satisfy the need of being a part of a group, particularly when they lack a sense of belonging to their social group at schools or universities.

Two U&G factors, maintaining interpersonal interconnectivity and entertainment value are found to be significant in explaining young users' engagement with Facebook in Brazil. Both factors reinforce the virtual community aspect of Facebook for Brazilian young users. They are keen to stay in touch with their friends and have fun via Facebook. Purposive value, self-discovery and social enhancement were not significant to explain user engagement. Furthermore, our results show that social presence did not significantly influence Facebook engagement among young Brazilian users. This is in direct contrast to Cheung et al.'s (2011) finding that social presence had the strongest impact on intention to use Facebook in Hong Kong. Moreover, our results of social influence factors (subjective norm and social identity) are also very different from those of Cheung et al. (2011) who found that group norm has a significant influence on intention to use Facebook, while social identity had no significance. In contrast, group norm did not have a significant effect for Facebook user engagement in Brazil. In Brazil, users do not seem to look for similarity of their values with their friends in Facebook. This finding indicates that there is no adoption of common self-guides for meeting idealized goals shared with others (Dholakia et al., 2004). The difference in Individualism scores between Hong Kong and Brazil (25:38, The Hofstede Centre, 2015) indicates that Hong Kong tends to have a more collectivistic culture than Brazil. This may help to partially explain that group norm, social enhancement and social presence are significant for Hong Kong students while not for the Brazilian young users. Overall, our model accounts for 42% of the variation in user engagement with Facebook. Comparing this with the 28% found in Cheung et al.'s (2011) study, the research model in our study shows a significant improvement in its explanatory power. Our results show unique factors of Brazilian Facebook user engagement from previous studies, reflecting the specificity of cultural contexts.

5.2. Implications for practice

Our findings have practical applications for companies wishing to connect and to engage young people in Brazil on Facebook. First, this study has revealed that subjective norm is the main driver of user engagement with Facebook. Managers of brand fan pages should attempt to find out the opinions of the leading members in

the reference group of their target customers. They could select and highlight the posts from leading members in the firm's Facebook updates to promote user engagement. As suggested by McCorkindale (2010), companies should not just use Facebook for information dissemination, but to engage and build relationships with stakeholders.

Second, given social identity is a significant driver, firms would need to convert their Facebook pages into a desirable online brand community for young Brazilian customers, by fostering their sense of belonging to this community. SNS marketing planners will need to consider manners to promote users to share their social lives and opinions on their brand fan-pages. They can also perform and display actions related to the social identity of the group reinforcing its belonging and identification. For example, a Facebook fan-page of adventure sports would highlight an environmental Brazilian concern activity to reinforce the social identity of the group.

Third, as can be expected, this study further confirms entertainment value is a driver for young Brazilian user engagement with Facebook. Information managers need to design and deliver interesting, fun, and innovative content and activities on Facebook to match their young fans' entertainment requirements. Since one the reasons users use Facebook is recreation, posting pictures and videos is a good alternative for maintaining an engaged audience of followers. However, more research is needed regarding this confirmation because what is considered interesting and fun is relative. What is considered interesting or fun in one culture, may not be in another one.

Fourth, this study reveals that maintaining interpersonal interconnectivity is another U&G value that drives user engagement in Brazil. This suggests that Facebook is an ideal medium to support a firm's customer relationship and maintain a connection with its existing and former customers for a long duration of time. The objective must be to keep them engaged, even when they grow older and no longer require the company's services. On Facebook, posts from the company must provide useful information and entertainment as well as maintain a relationship, avoiding explicit sales and messages that were not requested. Another option would be to create games, keeping the individual connected with the brand.

5.3. Limitations and further studies

There are several limitations that need to be considered when interpreting the results. First, the study findings are limited to one specific SNS, Facebook. Further investigation of other different SNS will generate interesting results. Second, the sampling method used is a non-probabilistic one. Nevertheless we have participants from all regions of Brazil and the sample profile is similar to that of young people in Brazil. Further research could include other groups and to verify the impact of different demographic and cultural factors. Finally, although the research model is appropriate for this

study's purpose, future research could improve the model by including moderating and mediating factors influencing user engagement with SNS, such as life satisfaction, personality traits and privacy concern, and adding behavioural variables as the consequences of SNS user engagement.

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Appendix A. Cross-loading

	EV	GN	INT	MII	PV	SD	SE	SI	SN	SP
EV1	0.86	0.34	0.35	0.58	0.41	0.34	0.36	0.43	0.30	0.43
EV3	0.84	0.27	0.29	0.44	0.38	0.40	0.38	0.31	0.29	0.46
EV4	0.83	0.22	0.29	0.44	0.32	0.30	0.35	0.28	0.21	0.34
GN1	0.31	0.93	0.31	0.38	0.39	0.29	0.22	0.49	0.26	0.31
GN2	0.32	0.95	0.37	0.40	0.38	0.25	0.19	0.55	0.32	0.33
INT1	0.32	0.36	0.94	0.40	0.34	0.22	0.12	0.49	0.5	0.30
INT2	0.36	0.33	0.94	0.42	0.36	0.23	0.12	0.47	0.52	0.30
MII1	0.48	0.33	0.32	0.83	0.56	0.55	0.47	0.38	0.32	0.49
MII2	0.52	0.38	0.43	0.91	0.46	0.34	0.25	0.49	0.37	0.42
PV1	0.32	0.31	0.31	0.47	0.72	0.36	0.23	0.40	0.33	0.35
PV3	0.34	0.31	0.31	0.46	0.79	0.42	0.34	0.35	0.25	0.37
PV4	0.35	0.36	0.32	0.49	0.84	0.43	0.31	0.39	0.32	0.43
PV5	0.33	0.33	0.29	0.46	0.75	0.43	0.27	0.40	0.29	0.40
PV6	0.32	0.28	0.29	0.41	0.69	0.37	0.28	0.32	0.28	0.38
SD1a	0.31	0.24	0.15	0.36	0.57	0.87	0.54	0.23	0.22	0.45
SD1b	0.43	0.26	0.27	0.50	0.56	0.89	0.50	0.29	0.27	0.41
SD2	0.33	0.25	0.18	0.38	0.57	0.88	0.54	0.24	0.25	0.47
SE1	0.41	0.21	0.13	0.39	0.46	0.55	0.96	0.16	0.16	0.42
SE2	0.44	0.20	0.10	0.36	0.46	0.59	0.94	0.15	0.16	0.45
SI1	0.30	0.49	0.38	0.40	0.39	0.27	0.18	0.73	0.38	0.34
SI2	0.36	0.44	0.39	0.44	0.38	0.27	0.18	0.78	0.35	0.31
SI3	0.36	0.42	0.46	0.45	0.38	0.24	0.11	0.86	0.39	0.34
SI4	0.36	0.50	0.47	0.43	0.38	0.23	0.13	0.88	0.38	0.36
SI5	0.31	0.47	0.43	0.42	0.38	0.24	0.14	0.87	0.38	0.32
SI6	0.30	0.47	0.42	0.40	0.38	0.24	0.11	0.86	0.38	0.31
SN1	0.29	0.25	0.48	0.34	0.31	0.27	0.16	0.38	0.91	0.31
SN2	0.29	0.32	0.51	0.40	0.36	0.25	0.15	0.44	0.92	0.32
SP1	0.44	0.30	0.24	0.42	0.43	0.42	0.42	0.33	0.27	0.83
SP3	0.45	0.32	0.36	0.51	0.44	0.36	0.26	0.42	0.36	0.84
SP4	0.31	0.22	0.15	0.31	0.38	0.43	0.44	0.20	0.19	0.76
SP5	0.34	0.25	0.19	0.34	0.4	0.43	0.39	0.26	0.26	0.80

Notes: Boldface numbers are loadings of indicators to their own construct; other numbers are the cross loadings.

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